ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

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DECLARATION

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ABSTRACT

This research presents an interactive human-robot collaborative learning platform designed to enhance cognitive and verbal skills in special needs students while remaining applicable to typical learners. Traditional teaching methods in special education often rely on static and repetitive approaches, limiting engagement and adaptability to individual learning needs. To address these challenges, the proposed system integrates computer vision, robotics, and speech processing to create a multimodal learning environment. A custom dataset of colored objects was developed and used to train a YOLO-based object detection model for real-time pattern recognition and object sorting tasks. Additionally, a speech interaction module utilizing Google Speech-to-Text and a logistic regression classifier enables verbal learning through sequence recognition and feedback. A 6-degree-of-freedom robotic arm, controlled via Arduino, provides physical demonstrations to reinforce both visual and verbal learning activities. The system was designed based on discussions with SERD Kelaniya Special Needs School and special education experts, ensuring practical relevance and usability. Experimental observations indicate improved engagement, interactive learning capability, and potential for real-world deployment. The study demonstrates the effectiveness of human-robot collaboration in developing inclusive, adaptive, and scalable learning solutions for special education.

Keywords: Human-Robot Collaboration, Special Needs Education, Object Detection, YOLO, Speech Recognition, Multimodal Learning

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LIST OF ABBREVIATIONS

Abbreviation	Description
6-DOF	6-DOF Robot Arm
API	Application Programming Interface
CNN	Convolutional Neural Network
HRC	Human-Robot Collaboration
IoU	Intersection over Union
mAP	Mean Average Precision
ML	Machine Learning
NLP	Natural Language Processing
YOLO	You Only Look Once – Machine Learning Model

CHAPTER 1: INTRODUCTION

1.1 Background Literature

Human-robot collaboration (HRC) has emerged as a pivotal domain of inquiry in robotics and artificial intelligence, propelled by the increasing demand for intelligent systems capable of functioning efficiently in human-centric settings. HRC is different from traditional industrial automation systems because it focuses on getting robots to work with people in the same space. Traditional industrial automation systems work on their own and follow pre-programmed instructions. This model focuses on being flexible, being able to talk to people in real time, and working together to get things done. This way, robots can help people instead of replacing them.

Collaborative robots have become really good at working with people because of developments in machine learning and computer vision. These robots can look around recognize things understand what people are doing and react in a way that makes sense. This means collaborative robots can be used in different areas, such as making things helping people get better and teaching. Collaborative robots are especially good at doing things that need to be done in a way even when things are changing around them.

In the field of education, people are getting more interested in using machines to help teachers and students learn. Educational robots are a way to get students involved and excited about learning. Robots can help students learn by giving them feedback away showing them things in a physical way and getting them to participate. This is really helpful when students have needs and learn in different ways. Educational robots can make learning fun and interactive, for all students.

Special needs education presents unique challenges that require personalized and adaptive teaching approaches. Students with cognitive, developmental, or communication difficulties often benefit from multisensory learning methods that combine visual, auditory, and physical interaction. However, many traditional teaching methods rely on static materials, repetitive exercises, and teacher-centered instruction. These approaches may not fully address individual learning differences or maintain sustained student engagement.

Recent studies have shown that interactive technologies, particularly robot-assisted learning systems, can significantly enhance engagement, motivation, and learning outcomes among special needs students. By providing consistent interaction, real-time feedback, and structured guidance, robots can support the development of cognitive and communication skills. Furthermore, the use of multimodal learning approaches - where students interact through multiple sensory channels - has been proven to improve retention and understanding.

Improvements in computer vision, especially deep learning-based object detection models like YOLO, have made it possible to recognize objects in real time in changing environments. These technologies help systems understand and make sense of visual information correctly, which is the basis for interactive applications. In the same way, advances in speech recognition and natural language processing have made it easier for people to talk to machines by letting them use natural speech.

Even with these improvements, we still need to combine different technologies into a single system that can support interactive learning well. A complete learning environment that meets the needs of all students can be made by combining vision-based perception, robotic manipulation, and speech interaction. Integration like this is especially useful in special education, where personalized feedback, engagement, and adaptability are all important for learning to be effective.

The idea of people and robots working together comes from the early days of industrial automation, when machines were made to do the same tasks over and over again with great accuracy and consistency. Over time, the focus changed from pure automation to collaborative interaction, where robots and people could work together in the same space and share responsibilities in a way that worked well for both. This change happened because people realized that when human thinking skills and robot physical skills work together well, they can do things that neither could do on their own. HRC is based on ideas from many fields, such as cognitive science, robotics engineering, machine learning, and human-computer interaction.

In the context of education, the rapid evolution of digital technologies has transformed traditional classroom environments into interactive and technology-enhanced learning

spaces. The integration of artificial intelligence, robotics, and multimedia tools has opened new possibilities for personalized learning, adaptive assessment, and inclusive education. These developments are particularly significant for students with special needs, who require individualized instructional strategies and additional support to achieve their academic and developmental goals. Educational technology has evolved from simple computer-aided instruction to sophisticated intelligent tutoring systems that can monitor learner progress, adapt content difficulty, and provide targeted feedback.

Neuroscientific research has further reinforced the value of multimodal and experiential learning approaches. Studies in educational neuroscience indicate that learning is most effective when it engages multiple cognitive systems simultaneously, including visual processing, auditory comprehension, and motor skill activation. This neurological basis provides strong support for the use of interactive robotic systems in educational settings, as these systems naturally engage multiple sensory and cognitive pathways. For students with special needs, particularly those with autism spectrum disorder (ASD), intellectual disabilities, or attention-deficit hyperactivity disorder (ADHD), multisensory learning approaches can significantly enhance attention, comprehension, and information retention.

The field of assistive technology has also played a big role in making learning spaces more welcoming to everyone. Assistive technologies include many different kinds of tools and systems that help people with disabilities do things that would be hard or impossible for them to do on their own. In schools, assistive technologies include augmentative and alternative communication (AAC) devices, screen readers, adaptive input devices, and more and more, interactive robotic systems. The merging of assistive technology and educational robotics is a promising new area for making learning spaces that are both fun and easy for all students to use.

Sri Lanka, as a developing nation, faces particular challenges in providing high-quality special needs education. Limited resources, trained personnel, and specialized equipment mean that many students with special needs do not receive the individualized attention and support they require. SERD Kelaniya Special Needs School is one example of an institution striving to address these challenges.

Developing cost-effective, scalable, and locally applicable technological solutions is therefore not only academically significant but also socially impactful. The proposed system aims to bridge this gap by providing an accessible platform that can enhance learning outcomes without requiring extensive infrastructure or highly specialized personnel.

1.2 Research Gap

Even if educational technology and human-robot collaboration have advanced significantly, there are still a number of issues, especially when it comes to special needs schooling. Without successfully combining several modalities into a coherent learning platform, the majority of current systems often concentrate on a single method of interaction, such as speech-based communication or visual recognition.

Furthermore, a lot of robotic learning systems don't have real-time adaptability, which means they don't change in response to changes in the environment or student performance. As a result, their ability to facilitate customized learning experiences is diminished. Additionally, even while some systems have robotic elements, they frequently fail to provide meaningful physical engagement to reinforce learning concepts.

The lack of usability and accessibility of current technologies is another significant drawback. Many sophisticated robotic systems are not suitable for general usage in schools, especially in special education settings, because they are costly, complicated, and require specialized infrastructure. Additionally, little study has been done explicitly on the use of human-robot cooperation to concurrently improve verbal and cognitive abilities.

The development of an affordable, interactive, multimodal learning system that combines speech-based communication, physical robotic interaction, and vision-based object identification is thus clearly lacking. In an inclusive learning environment, such a system should be able to support both pattern recognition and verbal skill development, provide real-time feedback, and adjust to user performance.

The comprehensive integration of vision, voice, and physical robotic interaction within a unified learning platform has not received enough attention, despite the fact that

separate components of educational robotics systems have been thoroughly examined. There is a large knowledge gap about how the combination of all three modalities affects learning outcomes because the majority of research concentrates on one or two modalities. In the context of special needs education, where the multimodal approach is not only advantageous but possibly necessary for successful learning, this disparity is especially noticeable.

Furthermore, the majority of existing studies on educational robotics have been conducted in developed country contexts, with limited applicability to resource-constrained environments in developing nations such as Sri Lanka. The hardware requirements, cost structures, and technical expertise assumed in many international studies are often misaligned with the realities of local educational institutions. This highlights the need for research that develops and validates systems under realistic local conditions, using affordable and accessible components such as Arduino microcontrollers and open-source software frameworks.

Another drawback is the evaluation techniques employed in earlier research. Without thoroughly evaluating engagement levels, instructor usability, or the sustainability of real-world implementation, many current systems are assessed using limited performance measures like task completion rates or accuracy scores. To fully capture the educational impact of interactive robotic learning systems, a more comprehensive evaluation approach is required. By integrating system performance measures, observational analysis, and stakeholder feedback into its evaluation methodology, this study fills this gap.

Additionally, there is a notable absence of research that simultaneously addresses both cognitive skill development (such as pattern recognition and object identification) and verbal communication skill development within the same platform. These two areas are frequently treated in isolation, despite the evidence that they are developmentally interconnected. For special needs students, integrating both skill areas into a single interactive system offers the potential for more comprehensive and efficient learning support. This integrated approach constitutes a key innovation of the proposed system.

1.3 Research Problem

Interactivity, flexibility, and real-time feedback mechanisms all crucial for successful learning among kids with a range of needs are frequently absent from traditional special education teaching methods. Reduced engagement, slower skill development, and trouble keeping students' attention can result from these constraints. Furthermore, because they either concentrate on discrete features or fail to offer an integrated learning experience, current technological solutions fall short of meeting the demands of students with special needs. The possibility of developing interesting and productive learning environments is limited by the lack of systems that integrate verbal, physical, and visual engagement.

Therefore, the research problem addressed in this study is the need to design and develop an interactive human-robot collaborative learning system that can enhance pattern recognition and verbal communication skills through a multimodal, adaptive, and user-centered approach.

Another challenge that the research problem faces arises from the need for a perfect balance between sophistication and practical application of technology. A system that is sophisticated but not usable by educators and students will not serve any useful purpose while a usable system that is not intelligent enough to respond to unique requirements of learners cannot help either. Therefore, the research problem calls for a solution that is the perfect combination of technical ability, user-friendly features, and educational benefits.

There is no question that the economic aspect of the research problem has to be considered too due to the fact that special needs schools in Sri Lanka, similar to those in any other developing country, operate on a restricted budget. To put it differently, the system will need to incorporate both hardware and software solutions that are cheap. This was achieved by opting for Arduino-based robots, machine learning algorithms developed by utilizing open-source resources, as well as cloud-based speech recognition systems. Pedagogically speaking, the research problem also includes issues related to the compatibility of technological advances with current special education theory. As mentioned before, special needs education is based on

principles such as UDL. Thus, the system being developed cannot simply function as a technology; it has to serve as an educational tool as well.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to develop an interactive human-robot collaborative learning system that enhances pattern recognition and verbal communication skills in special needs students through a multimodal and adaptive learning environment.

1.4.2 Specific Objectives

To achieve the main objective, this research focuses on several specific objectives, as follows:

- To design and develop a real-time object detection system using deep learning techniques, specifically the YOLO algorithm, to accurately identify predefined objects within the learning environment.
- To create a custom dataset and use it to train and evaluate the detection model under realistic conditions.
- To implement a robotic interaction system using a 6-degree-of-freedom robotic arm controlled via an Arduino microcontroller, enabling physical demonstration of learning patterns.
- To develop a pattern-based learning mechanism to allow students to observe and replicate object sequences, thereby enhancing cognitive and pattern recognition skills.
- To integrate a speech-based interaction module that converts visual patterns into verbal instructions, supporting language development.
- To apply a logistic regression model for voice command analysis, enabling the system to determine the accuracy of spoken inputs.

- To establish a multimodal learning framework that combines visual, physical, and verbal interaction, and to assess its effectiveness in improving student engagement, cognitive abilities, and verbal communication skills.

Together, the objectives constitute a research agenda that covers all the research gaps and needs arising out of the study. All the objectives will be directed towards achieving the ultimate goal of establishing a sound system that is technologically feasible and educationally useful. The objectives have been developed after extensive consideration of technological feasibility and educational benefits through consultation with educators and literature review.

The study is based on the assumption that the constructivist theory is the best approach to learning since it assumes that people learn by interacting with the environment around them. This assumption is embedded in the design of the system, whereby the student does not simply take the information presented; rather, he/she learns actively by observing, reacting, and receiving feedback within an interactive learning environment. The constructivist theory is especially appropriate for special education programs because the learner acquires information more effectively when actively involved.

CHAPTER 2: LITERATURE REVIEW

2.1 Human-Robot Collaboration and Adaptive Systems

From a rigid and predetermined system of automated processes, human-robot collaboration (HRC) has advanced into an intelligent and adaptive process that allows a robot to interact with a human being and operate in the environment. The earlier version of such technologies, like collaborative robots, was a way of using robots in assisting human beings through safety and precision without considering behavioral aspects, which set the basis for collaboration between human beings and robots [6]. Currently, with the development of AI, modern HRC technologies include perception, learning, and decision-making [1], [3], [11].

One important aspect of HRC is the ability to learn and respond according to human behavior. Learning methods and adaptive control mechanisms help in making the robot able to anticipate human action and then adapt its own actions to support humans in their tasks [5], [3]. Adaptive pattern recognition helps robots in learning about the environment and then adapting based on the context by analyzing their environment and taking suitable actions [4]. The underlying idea is from the field of AI, whereby intelligent machines sense their environment and take appropriate action in order to achieve a specific goal [28]. In addition, the idea of human-centric AI is also important in designing such intelligent machines [29].

The invention of collaborative robots or collaborative robots represents a major breakthrough in the field of HRC. As opposed to conventional robots that are used in an isolated and strictly controlled environment, collaborative robots are created to coexist and interact with humans in safe and efficient manner. With the help of sophisticated sensors, force control and collision detection systems, collaborative robots are capable of reacting properly to human interaction and behavior. Due to their safe design, collaborative robots have become the perfect solution for implementation in places where their usage was not feasible before, such as hospitals and schools.

Machine learning technology has had a revolutionary impact on improving the adaptability of HRC systems. In the case of reinforcement learning, robots are taught appropriate behavior based on interacting with the environment through trial and error

without being explicitly programmed for all possible cases [10]. This type of machine learning technique has immense importance in dynamic and unpredictable environments like the classroom setting where robots need to adapt according to student behavior and environmental changes. Transfer learning techniques have sped up the process of building adaptive HRC systems by allowing training data sets from general purposes to be adapted for specialized applications.

There has been an increase in research into the social aspect of the HRCs in the past few years. Socially aware robots that can detect and react to human social behaviors such as body language and tone of voice show great potential when used in education and therapy. Studies on social robotics have found that people including autistic patients are more likely to interact with social robots than any other method of instruction or therapy from humans. The reason for this increased social interaction with robots rather than human beings is that social robots do not judge their interlocutors and are extremely patient and therefore less intimidating.

Another aspect of HRC systems that needs consideration is shared autonomy. According to shared autonomy schemes, the allocation of control of a specific task depends on how capable or willing each individual is to take it over. If we consider this idea within the context of education, there could be systems that operate by initially letting the robot assume the role of leading through a task, with control shifting from the robot to the user progressively as the user gains competence. Such approaches in teaching are similar to theories like the zone of proximal development according to Vygotsky, wherein learning is most effective when conducted beyond the level of independence.

2.2 Vision-Based Perception and Object Detection

Perception based on vision plays a key part in achieving the goal of real-time interaction within HRC applications. Classic approaches to computer vision made use of manually crafted features in conjunction with a controlled environment. With the development of deep learning, the task of detecting objects and recognizing them has been greatly enhanced. The application of deep learning in the field of computer vision was first achieved via convolutional neural networks (CNN), as seen in the case of image classification problems [15]. This was followed by ResNet, which enabled deeper layers and more accurate feature extraction [16]. The next big breakthrough came through region-based CNN, namely, the Fast and Faster R-CNN [17], [18].

Among the current detection systems, YOLO (You Only Look Once) detection system is one that has gained popularity due to its ability to detect objects in just one pass [7], [8]. The detection of an object at once makes the system very effective and efficient for use in interactive systems, which require low latency. Apart from detection systems, some pose estimation systems like OpenPose have been developed, and they aid in detecting human body movements [9].

It is observed that there has been a constant progression in object detection algorithms, which includes speed, precision, and generality. Rule-based systems that made use of predefined features like edges, corners, and histograms based on colors have certain limitations since they tend to be affected by environmental changes and cannot be generalized for other objects. However, with the emergence of deep learning and CNNs, this approach has revolutionized the field of object detection. With CNNs, there has been a paradigm shift from predefined features to learned features.

The development of the YOLO object detection models is among the most influential developments to emerge from real-time computer vision research. The core idea behind the YOLO algorithm is the transformation of the object detection process into a regression task that predicts bounding boxes and classes for all objects within an image at once. Since the entire process is conducted during a single pass, the separation between proposal and classification phases is not required anymore, and therefore the

speed is tremendously increased. Future versions of YOLO made several improvements to the architecture of the model.

Transfer learning has emerged as a vital strategy in applying deep learning-based vision systems in real-world applications. With pre-trained weights from massive datasets like ImageNet or COCO, scientists can save on the amount of task-oriented data necessary to obtain a high level of accuracy for object detection. This is especially useful when developing vision solutions for niche industries like educational robotics where acquiring and labeling vast amounts of data can be too expensive. The current research utilized transfer learning as it used pre-trained YOLO to perform detection on the custom learning object dataset.

Another equally important aspect that can be mentioned in relation to improving the reliability and performance of deep learning vision models is data augmentation. Data augmentation procedures include random image cropping, flipping, rotation, brightness adjustment, and color augmentation, among others. These methods help increase the size of the training set and enhance its diversity through creating artificial variations from existing training images. Consequently, this allows the model to develop the ability to extract information relevant to object classification despite such irrelevant alterations as changes in the image's orientation, color intensity, or even illumination. In this case, data augmentation was used to train the detector so that it works properly in any classroom environment.

Apart from object detection, the latest progress in semantic and instance segmentation has created opportunities in the field of interactive computer vision beyond the scope of object detection. Through semantic and instance segmentation, machine learning models will not only be able to recognize the position of objects but will also be able to differentiate between individual objects within the same class of objects. Although more computationally intensive than object detection, there has been an increasing effort towards making such models more efficient and thus more realistic. Future versions of the proposed system may take advantage of such advances.

2.3 Speech Interaction and Machine Learning Approaches

Speech systems have emerged as another crucial way of interacting with robots, apart from vision systems. Natural language processing, along with speech recognition, has made it possible to communicate with robots in a simple way. The fundamentals of speech and language processing have formed the foundation of transforming verbal commands into computational representations [19]. Various machine learning algorithms, such as logistic regression and other supervised learning methods, have been extensively employed to categorize data sets, especially in evaluating the response to spoken commands [20], [21]. Deep recurrent neural networks are a more sophisticated means of recognizing speech through modeling temporal data [22].

Speech recognition technology has undergone significant evolution through the years, starting from initial stages where it was limited to recognizing single words chosen from a constrained vocabulary set. The development of Hidden Markov Models (HMMs) towards the end of the 1980s marked an important turning point with the ability to incorporate continuous speech recognition into their functionality. The combination of GMMs with HMMs brought about even better performance in recognizing the variability that exists in the production of phonemes and speaker characteristics. It was until recently that deep learning replaced these statistical models as the dominant methodology.

The emergence of deep neural networks (DNNs) as acoustic models in the early 2010s marked a pivotal transition in speech recognition technology. DNN-based acoustic models consistently outperformed GMM-HMM systems across standard benchmark datasets, demonstrating the superior ability of deep learning to capture complex acoustic patterns. The subsequent development of end-to-end speech recognition architectures, including Connectionist Temporal Classification (CTC) and attention-based sequence-to-sequence models, further simplified the speech recognition pipeline by eliminating the need for explicit phoneme alignment during training. These architectures learn to map raw audio features directly to character or word sequences, achieving state-of-the-art performance with reduced system complexity.

Whisper by OpenAI, which is used in the suggested system for converting speech into text, is considered to be one of the most important breakthroughs in the field of speech recognition technology. Being trained on a huge database of multilingual audio files, Whisper has proven its exceptional ability to cope with such difficulties as acoustic variations and background noises as well as the different accents of speakers. This ability of whisper to understand speech uttered by speakers who pronounce sounds differently and have various voices will be extremely beneficial in the educational setting of the suggested system, in which learners with disabilities will participate.

The natural language processing (NLP) methods enhance the process of speech recognition because they allow the system to comprehend and analyze the semantics of the transcribed speech. In our system, tokenization will help us to split transcriptions of students' answers into tokens based on objects, which will then be systematically compared with the expected sequences. Advanced NLP approaches like named entity recognition and semantic similarity can be considered for future versions of the system in order to evaluate the students' answers more flexibly. For instance, semantic similarity will detect whether the answer reflects the same idea as the expected sequence.

The incorporation of speech interaction in the learning interface represents an emerging paradigm shift towards natural user interfaces (NUI) in educational technology systems. Natural user interfaces, which comprise speech, gestures, and touch-based interactions, seek to minimize the mental and physical challenges posed by conventional input methods like keyboards and pointing devices. In the case of learners with motor difficulties or low literacy levels, the use of speech interaction may offer a more user-friendly method of interacting with learning materials. Studies carried out on natural user interfaces in educational settings have revealed that natural interaction methods enhance student motivation and lead to better learning results, especially for young children and disabled learners.

2.4 Educational Robotics and Collaborative Learning

Another interesting application field for HRC is that of educational robotics. According to recent research, educational robotics can play an important role in increasing students' motivation, engagement and learning completion rates through the use of interactive learning experiences [23]. Robotic assistance can serve as a facilitator during the learning process, ensuring constant interaction and provision of feedback. It was also found that robots can be quite helpful in the development of special needs children as they help in improving children's communicative and cognitive abilities [26], [27]. Interactive learning processes that imply direct teaching by or interaction with robots lead to improved learning outcomes [25].

The problems associated with breakdowns in interaction in the area of human and robotic interaction in education were also considered. These research works pointed to the need to design more robust and flexible systems capable of maintaining engagement even in case of interaction failures [24]. Multimodal learning systems utilizing various channels of interaction, such as visual, physical and verbal ones, have proved to be especially effective in terms of engaging special needs children in the learning process.

In order to prove the relevance of educational robotics to the modern classroom, it is important to consider the theoretical foundation behind their effectiveness. First, constructivism is the most appropriate theory to explain the significance of educational robotics. According to Piaget and Vygotsky, this theory emphasizes the importance of hands-on experience in which people actively build up their knowledge by interacting with their environment. Through the use of a robotic system, one creates learning experiences that are constructivist as such. In this case, learners should actively observe, interpret, and react to the robot's actions thus increasing their engagement.

Another important theory that can be used to discuss the impact of robotic interaction on the learning process is embodied cognition. According to the researchers associated with this theory, cognitive processes are directly related to the human body and its interaction with the surrounding environment. In this way, using a robotic system to provide an example of a particular pattern will engage the learner's embodied cognition

because the process includes some physical actions that enhance learning and improve memory.

The studies specifically investigating the role of robots in assisting ASD children have proved especially convincing. Children with ASD who experience difficulties with human interactions often display lower levels of social anxiety and greater engagement while engaging with social robots. Social robots offer a socially predictable, reliable, and emotionally unthreatening social entity that many children with ASD find less overwhelming than interactions with humans. Lower social anxiety may help create an optimal emotional climate for learning by allowing students to concentrate better on their learning activities. There were reports from several studies that communication skills, social engagement, and ability to focus had improved after participating in robot-assisted intervention programs.

One approach, which has proven especially productive for educational robotics development, is positioning robots as companions, not instructors or tools. According to this approach, robots serve as learning companions that learn with the child, make mistakes, show confusion, and possess a growth-oriented mindset. It was found that interacting with robots positioned as learning companions led to higher levels of motivation, perseverance, and metacognitive reflection than those exhibited when students interacted with instructor-oriented robots. This finding has important implications for the design of the proposed system, where the robot's role as a demonstrator and interactive partner positions it as a collaborative learner rather than a didactic teacher.

Feedback is an important aspect within education in robotics, which requires special consideration. Feedback is an essential part of learning, and robotics presents many possibilities to give the learner immediate and structured feedback. Human teachers cannot be consistent in giving feedback as they have different styles depending on individual qualities. However, robotic systems allow one to provide precisely measured feedback for definite actions performed by the learner. This consistency proves to be especially beneficial for children who have some special learning needs, as they need consistent structure and stability in the process of learning. Both visual and auditory types of feedback will be applied in this system.

2.5 Collaborative System Design, Limitations, and Research Direction

From a standpoint of system design, collaboration in robotics also includes task allocation and motion planning. Collaborative sorting and task allocation in robots have proven the importance of dynamic task allocation based on behavior of humans and their surroundings [14]. Frameworks for decision making, including Markov models and predictive planning, enable robots to predict human actions and increase the efficiency of cooperation [13]. Moreover, creating predictable motions allows robots to become easily understandable and reliable partners in interaction with humans [12].

Notwithstanding the considerable progress achieved, most of human-robotic interaction solutions still cannot be used for practical application on a large scale. Effective implementations require not only advanced algorithms but also affordable hardware and simple interfaces for end-users. Systems implemented using Arduino can serve this purpose well as they allow for the combination of hardware with sophisticated machine learning solutions [6]. Moreover, in line with recent trends, contemporary AI systems take into account the importance of accessibility and usability of their interfaces [28][29].

The literature provides strong evidence of the fast-paced evolution of HRC technologies through the development of machine learning, computer vision, and human-centered design methods. Object detection in real time, speech interaction, and robotic manipulation form an excellent basis for building interactive and adaptive learning systems. These developments are applied in the current research by designing a multimodal human-robot collaborative platform aimed at improving special needs education. The incorporation of vision-based pattern recognition, speech-based interaction, and robotic demonstration makes the proposed solution unique and enhances its value for the field of inclusive intelligent technology.

The analysis of the literature suggests that the evolution of HRC technologies occurs in stages. Initially, researchers and practitioners focus on creating robots capable of performing one type of function efficiently, while more recently, attention is paid to building multimodal systems that unite multiple aspects of human-robot interactions

and perceptions. Thus, it can be argued that the development of effective and inclusive HRC technologies involves integrating various aspects of technological advancement rather than optimizing any particular element.

Furthermore, the review emphasizes the role of co-designing and participation of all stakeholders in the development of educational technology systems. Educational technology systems that lack significant feedback and participation from end-users, including teachers, students, and parents, are often unsuccessful in implementation and long-term use within educational institutions. Co-designing, which involves educators and students in the design and testing of the educational technology, is known to result in better usability, relevance, and effectiveness of the educational technologies. In this study, stakeholder engagement with SERD Kelaniya Special Needs School will be an integral part of the methodology used.

The ethical aspects of the development of educational technologies, especially those that involve special needs children, are gaining significance in the field of educational technology. Privacy issues, consent to collect and process information about the participants, biased algorithms, and the risk of substituting humans in education are some of the problems that should be considered when developing an educational robotics system. The proposed educational robotics system will address ethical concerns by conducting proper data collection and processing procedures, designing the robot as an additional educational resource, not a substitute for humans, and explaining the decisions made by the system.

CHAPTER 3: METHODOLOGY

3.1 System Overview

The current study addresses the design and development of a multimodal human-robot cooperative learning system suitable for use in special education contexts. This system utilizes advanced technologies such as computer vision, machine learning, speech processing, robotics, and web interfaces to facilitate an interactive learning environment where students, teachers, and an intelligent robot can interact in real-time.

The key objective of this system is to improve pattern recognition, object identification, and speech communication abilities through interactive learning processes. The methodology adopted involves following a systematic development process comprising data set generation, training machine learning models, integration of robotics, speech processing, system integration, and evaluation processes.

The overall system architecture can be conceptualized as three interconnected layers: the perception layer, the processing layer, and the interaction layer. The perception layer encompasses the hardware and software components responsible for capturing and interpreting information from the environment, including the camera for visual input and the microphone for audio input. The processing layer includes the machine learning models and algorithms that analyze the captured data to generate meaningful representations, including the YOLO-based object detector and the Whisper-based speech recognizer. The interaction layer encompasses the output components that communicate with the learner, including the robotic arm, the speech synthesizer, and the web-based teacher interface.

The system is designed according to a modular architecture that facilitates independent development, testing, and updating of each component. This modularity is a key design principle that enables the system to evolve over time as new technologies become available or educational requirements change. Each module communicates with others through well-defined interfaces, ensuring that changes to one module do not require extensive modifications to others. This design approach also facilitates troubleshooting and maintenance, as issues can be isolated to specific modules without affecting the functioning of the overall system.

The development process was iterative and focused on users, involving cycles of designing, prototyping, testing, and refining the system. Such processes have been widely used in Human-Computer Interaction studies and educational technology research due to the complexity of learning and its dependence on many factors. Feedback received during each cycle of implementation made it possible to enhance the software's effectiveness and user-friendliness gradually. The process allowed the research group to detect and solve some problems that may have arisen when developing a fuller version of the system later.

The flow of data within the system starts with the capturing of the video frame of the work area by the camera. This process is continuously monitored by the YOLO algorithm, which is responsible for object detection and classification in real time. Detected objects and their positions are utilized in creating structured learning patterns that are then conveyed to the robotic arm via Arduino. In addition, detected patterns are translated into audio instructions using text-to-speech synthesis and sent to the student through the audio output. Voice responses from the student are received via the microphone, analyzed by the Whisper speech recognition program, and tokenized before being classified with the use of logistic regression.

3.2 Requirement Analysis and Stakeholder Consultation

The requirement analysis was carried out by conducting interviews with the stakeholders of SERD Kelaniya Special Needs School and the professionals who practice in the field of special education. The interview sessions were specifically aimed at finding the limitations that existed within the traditional teaching methods used with special needs students.

The findings highlighted the necessity for a learning system that embodies the following characteristics:

- Interactive and engaging rather than static.
- Capable of providing real-time feedback.
- Supportive of multisensory learning (visual, physical, and verbal).
- Simple and adaptable for classroom environments.

Accordingly, the architecture of the system was developed taking into consideration these considerations in order to design the system in a way that would enable intelligent perception, robot interaction, and voice interaction in the same learning environment.

The requirement collection process comprised organized discussion and observation within the classrooms of SERD Kelaniya Special Needs School conducted throughout several sessions. During these sessions, there were direct observations regarding current classroom activities, interviews with special education teachers, and conversations with children who were undergoing lessons. Through such processes, significant information related to the difficulties faced by students of varying abilities, current teaching techniques, and other factors such as limited space were gained.

The following pedagogic requirements emerged from the interviews conducted among the teachers. To begin with, teachers pointed out the significance of retaining the attention of students. In particular, the attention span of most special needs learners is short, thus necessitating a switch of activities in order to ensure continued involvement. Based on this requirement, a system that offers various types of activities has been designed as opposed to the provision of repetitive and static learning programs. Secondly, teachers pointed out the significance of timely and clear feedback, especially since delay or ambiguous feedback does not work with special needs learners.

In terms of technical requirements, consultations with stakeholders indicated the following issues to consider. Firstly, the system should function properly in an average learning environment, taking into consideration possible variations in light conditions, noise levels, and the number of students involved in learning. In addition, the hardware required to use this learning solution should be reliable, secure, and relatively easy to maintain since any failure or need to conduct maintenance operations might hinder educational activities. Finally, the software needed for this program should be easy to use even by less technically savvy teachers. These constraints collectively defined the technical design parameters of the proposed system.

Accessibility came out as the most critical requirement in the process of consultations among the stakeholders. Children with special needs have different profiles in terms of their cognitive, motor, and communicational skills; therefore, the system should be accessible to children irrespective of their ability. This requirement necessitated the use of a multimodal interface in which students could interact with the system using visual perception, voice recognition, and physical gestures, depending on their capability. The system interface for teachers was designed bearing in mind their level of computer skills.

3.3 Dataset Creation and Preprocessing

A custom dataset was manually created for training the object detection model. The images used contain simple objects including color-coded balls, boxes, and balloons since they are suitable for use in pattern recognition in early learning.

Images of the objects were taken in real-life situations and included different lighting conditions, angles, distance, and background scenes. This approach is important in ensuring that the model is effective in actual classrooms rather than in laboratory settings.

Annotation involved labeling the object category through the use of bounding boxes, while the datasets created were split between the training and validation datasets after preprocessing to ensure uniformity for optimal performance of the model during the training process.

There were certain guiding principles behind the creation of the image dataset that include: firstly, the dataset should be diverse since it is important to represent the various situations under which the system will be employed to make predictions or decisions. This is important because images were captured using different settings, i.e., under different artificial and natural lighting, at different times of the day, and using different backgrounds depending on the environment in which they were captured. Secondly, variation of object positions was an important principle since this helped in challenging the model as it was trained.

The annotation process involved labeling each image using the Label image annotation tool, which provides a graphical interface for drawing bounding boxes around objects of interest and assigning class labels. For each image in the dataset, annotators drew tight bounding boxes around each visible object and assigned the appropriate class label from the predefined set of categories. Quality control was implemented through a review process in which a second annotator checked a random sample of annotations for accuracy and consistency, with any discrepancies resolved through discussion and re-annotation.

Dataset splits followed the classical approach with a ratio of 70/15/15 for training, validation, and test sets, respectively. Training data was utilized to tune up model parameters during the training process, whereas validation data served the purpose of training monitoring. In turn, test data served as holdout data for performance testing. To keep categories proportionate among all data types, stratified splits were employed. Data preprocessing included resizing of images according to YOLO model requirements, normalization of pixel values to values that would be expected by the YOLO model, and label format conversion to the YOLO model format.

During model training, image augmentations such as random horizontal flipping, random scale change, mosaic augmentation, and HSV color space jittering were used. As albumentations library is considered to provide efficient image augmentation tools for object detection, all augmentations described above were implemented using it.

The class distribution in the dataset was carefully monitored to ensure that the model was not biased toward more frequently occurring classes. Where class imbalances were detected, additional images were captured and annotated for underrepresented categories to restore balance. This attention to class balance is particularly important in the educational context, where the system must perform reliably across all object categories to support fair and effective learning experiences for all students.

3.4 Object Detection Using YOLO Model

3.4.1 Model Selection and Justification

The YOLO (You Only Look Once), which stands for You Only Look Once, was chosen as the main vision part because of its capability to detect objects in real time at extremely fast speeds. Unlike other algorithms that use a two-step approach for detecting objects, YOLO analyzes the whole image in one feedforward pass.

3.4.2 Training Process and Learning Mechanism

The YOLO model was trained using the custom dataset created for this research. During training, the model learns to detect object locations and classify them simultaneously by optimizing bounding box predictions and class probabilities. The training process enables the model to understand the following aspects:

- Object identity (class recognition).
- Spatial location (bounding box prediction).
- Multiple object detection in a single frame.

Through iterative learning, the model becomes capable of accurately identifying objects such as colored balls, boxes, and balloons in real time.

Transfer learning was performed by pre-training the model with weights that had been trained on the COCO dataset, consisting of more than 200,000 labeled images with 80 different objects. As such, transfer learning enables the model to utilize knowledge gained from the larger dataset to detect objects in our small dataset and reduces the need for large amounts of task-specific labeled images. In addition, pre-trained weights were fine-tuned using stochastic gradient descent optimization by adjusting parameters such as the learning rate, batch size, and training epochs to improve validation accuracy.

Grid search hyperparameter optimization was used to optimize parameters such as the learning rate, batch size, and augmentation factor. A learning rate scheduler was applied, consisting of warm-up training and cosine annealing scheduling of the learning rate. The benefit of this learning rate schedule over constant learning rates has been shown in several studies. Regularization, in the form of dropout and weight decay, was implemented to prevent overfitting.

The model's performance was monitored throughout training using the validation set metrics, including precision, recall, and mean average precision (mAP). Training curves were examined to detect signs of overfitting, such as a divergence between training and validation loss, or a plateau in validation performance. Early stopping was applied to halt training when validation performance ceased to improve, preventing overfitting to the training data. The final model checkpoint corresponding to the best validation mAP was saved for deployment in the integrated system.

The need for computational efficiency required by real-time inference has been taken into account in the design and deployment of the architecture. The YOLO model has been set up to run at a speed that is fast enough to ensure an interactive and smooth experience for students. Real-time frame rate performance has been ensured using techniques such as model quantization, which helps reduce memory usage and computation speed by lowering the precision of the parameters in the model, as well as using hardware acceleration in the form of the GPU that has been made available.

3.4.3 Real-Time Detection and Pattern Formation

After training, the YOLO model is deployed in real-time to detect objects placed on the workspace. The system continuously captures frames and processes them to identify all visible objects. The detected objects are then used to generate structured learning patterns. These patterns form the basis for robotic demonstrations and verbal instructions, transforming raw visual data into meaningful educational sequences.

3.5 Robotic System Design and Control Mechanism

3.5.1 Hardware Architecture

The physical interaction of the system takes place through the use of a 6 degree of freedom (DOF) robotic arm. This type of configuration gives flexibility to the movement process, thus allowing the robot to interact with the environment similar to human interactions. Control of the robotic arm takes place through the use of the Arduino microcontroller that works as the interface between the machine learning and physical aspects of the system.

3.5.2 Motion Execution and Pattern Demonstration

After the recognition of the objects and generation of a learning pattern by the YOLO model, the robotic arm performs predefined motion commands which illustrate the learned pattern physically through various actions such as pointing to an object, arranging objects, and handling objects. It performs these actions simultaneously with generation of the sequence to ensure synchronized actions between recognition of visual data and illustration of the learning pattern in real-time, thereby allowing students to learn and recognize patterns using their observation skills.

The 6-DOF robotic arm is preferred due to its ability to generate the number of degrees of freedom necessary for a wide range of natural movements, making it possible for a robot to perform interactions in an easy and understandable way by students. Each degree of freedom entails a rotation joint, and six rotation joints give enough movements in all three dimensions which can be covered by a human arm. Such configurability allows the robotic arm to reach any point within its reach envelope and in any orientation, which can allow it to perform actions such as pointing, picking up, moving, and sweeping.

The controller system implemented using the Arduino platform communicates with the host computer through a serial USB connection, receives commands for movement, and then processes these commands into servo motor signals. In our system, each servo motor controls one degree of freedom of the robot manipulator. The angle of the motor is calculated according to the generated PWM signal. Calibration was carried out to determine the exact relationship between the coordinates used by the software system and the angular positions of the servos.

The motion planning algorithm is responsible for generating smooth trajectories by interpolating between two states, namely the starting point and the end point of motion. By interpolating the motion, it becomes smoother and more understandable. It is important to note that the trajectory generation is performed by taking into account constraints on speed and acceleration so that the operation of servo motors does not go beyond their permissible values. Collision avoidance constraints are applied to prevent the arm from colliding with objects on the workspace or with the student's hands during interaction.

The physical environment in which the system works is carefully arranged to account for both the physical needs of the robotic arm, as well as the educational purpose of the system itself. The objects are positioned on a level surface that is adequately illuminated and within reach of the robot, while also being clearly visible to the camera and within reach of the arm's end effector. Safety considerations such as soft joint limits and force torque monitoring have been incorporated into the design process to ensure safe operation in the presence of students.

Beyond simply demonstrating the operation of the robotic arm, its movement serves the additional role of a physical form of communication. The robotic arm can be understood as communicating through actions such as the grasping of objects and their manipulation within the workspace. The use of such actions for the sake of communicating information regarding objects, sequences, and their placement is based on the theory that gestural communication can be an effective tool when teaching children with certain special needs. Studies on educational gesture processing indicate the value of teacher gestures in promoting understanding, and the designed movement of the arm seeks to replicate them.

3.5.3 Educational Role of the Robotic System

The robotic system acts as an intelligent teaching assistant that bridges digital learning with physical interaction. By demonstrating patterns physically, the robot enhances cognitive understanding, visual memory, and engagement. This approach is particularly beneficial for special needs students, as it supports kinesthetic and visual learning styles, improving comprehension and retention of learning concepts.

3.6 Speech Recognition and Verbal Learning Module

3.6.1 Speech-to-Text Processing Using Whisper

The speech interaction component is realized by utilizing the OpenAI Whisper model, which is an efficient speech recognition model that can effectively transcribe spoken language to text regardless of the acoustic environment. Once the sequence for learning is created, it is transformed into instructions spoken out aloud. Students answer by repeating the sequence in verbal form, and the spoken input is acquired using a microphone. The Whisper model will process the speech input to provide a transcribed output.

3.6.2 Text Processing and Tokenization

After transcription, the system processes the text output using tokenization techniques. The sentence is broken down into individual object-based tokens such as "red ball," "green box," and "blue balloon." This structured format enables the system to compare student responses with expected sequences at a granular level, improving accuracy in evaluation.

3.6.3 Response Classification Using Logistic Regression

The use of a logistic regression model will be employed in order to test the validity of the verbal answer of the student in accordance with the anticipated sequence of learning. The model categorizes the verbal answers according to accuracy of tokens and sequence. In case the answer is correct, then the next phase of learning begins.

3.6.4 Adaptive Feedback Mechanism

When incorrect responses are detected, the system provides immediate corrective feedback through multiple channels. The robotic arm re-demonstrates the correct sequence, and the system repeats the verbal instruction. This multimodal feedback loop ensures continuous learning improvement by reinforcing correct patterns through both auditory and visual channels.

The speech recognition pipeline in the proposed solution consists of several consecutive phases, which help enhance the effectiveness of the verbal response evaluation. The first phase consists of the audio recording performed by a high-quality unidirectional microphone aimed at reducing noise interference and capturing only the sound of the student's voice. Next, the pre-processing phase includes the noise reduction, silent frames elimination, and sound normalization procedures that facilitate more precise recognition with the Whisper model. Such pre-processing steps are crucial since they increase recognition quality in noisy environments where the system will work.

The next phase of processing concerns tokenization, which transforms the continuous stream of words generated by the speech recognition model into the set of discrete tokens representing objects that students are naming. The tokenization procedure relies on the use of several natural language processing methods, such as word segmenting,

stop words filtering, normalization of color terms, and object names matching. It is important to mention that the tokenizer should be able to accommodate various ways in which students may refer to objects, such as 'the red one' instead of 'red ball', or 'big blue thing' instead of 'blue balloon'. The tokenizer is designed to map these variant expressions to the correct canonical object tokens, improving the robustness of the evaluation process.

The logistic regression classifier used for verbal response evaluation was trained on a labeled dataset of student responses that were collected during the development and testing phases of the system. The training data includes both correct and incorrect responses, with the class label indicating whether the student's sequence of object tokens matches the expected learning pattern. Feature engineering for the classifier includes sequence alignment scores, token overlap ratios, order accuracy metrics, and length difference features, which together capture the key dimensions along which student responses can differ from the expected sequences. The classifier outputs a probability score for correct classification, with a threshold-based decision rule applied to generate binary correct or incorrect judgments.

This particular feature is aimed at making an adequate reaction depending on the type and amount of the mistake made. In case when the answer provided is almost accurate and there are some mistakes related to the swapping of tokens or the slight mispronunciation of the name of the object, the feedback tool suggests a corrective suggestion that will pinpoint only the error without having to repeat the whole sequence. If the answer is inaccurate, the feedback is going to be more detailed as the teacher will need to show again all the steps of the process. Thus, the process of feedback resembles greatly that of scaffolded teaching technique used by experienced special education teachers.

3.7 System Integration and Learning Workflow

The system integrates three core components: vision-based object detection, robotic execution, and speech-based interaction. The workflow begins with YOLO detecting objects in real time. Based on these detections, structured learning sequences are generated and communicated to students through both speech output and robotic demonstration.

Responses made by students are done verbally and are captured and processed using the speech recognition and classification components of the system. Evaluation of students' responses takes place on-the-fly as feedback is given to them instantly, thus enabling an adaptive learning process.

The integration of all these components together has posed several challenges in terms of coordination of actions at certain points. For instance, the YOLO object detection algorithm continuously runs in a separate thread where it captures and analyses video images of the objects detected to update their location at all times. The robotic arm controller also gets motion requests from the control program to move the arm in a desired direction but does so asynchronously as it sends messages back to the control loop to notify about completion of the command. Finally, the speech recognition system also runs in another thread which waits until the audio input goes above a threshold before initiating the recognition task.

This workflow of the learning process session is represented using the state machine approach with explicitly specified states and events that trigger transitions between those states. Those states are: initialization; pattern generation; demonstration of a task using a robot; verbal instruction delivery; gathering of the student response; assessment of the response; delivering feedback; and terminating the session. State changes occur when particular events take place, like termination of a robotic motion sequence, termination of the student's verbal response, or triggering the timeout timer. Handling errors and recovering from failures in case of their occurrence is an important aspect of any software solution. In case of failure to detect any objects in the workspace, the system will generate the verbal prompt requesting a student to make sure that all necessary objects are present within the field of view of the camera. In case of failure to transcribe a student response by the speech recognition module, a

similar request will be made by means of a verbal prompt. In case of encountering any obstacles during a robot's motion sequence, this motion will be stopped and an appropriate warning will be generated.

3.8 Web-Based Interface and Teacher Support System

A web-based user interface was designed to assist teachers in the management and supervision of learning activities. It provides a systematic teacher guide and allows real-time observation of students' interaction. The teachers will be able to monitor students' performance, receive learning guidelines, and conduct live lessons within the web-based interface. Consequently, it ensures seamless integration of the system in classrooms while involving teachers in the process.

The web-based teacher interface was designed using responsive design techniques to accommodate various device platforms, such as desktops, tablets, and mobile phones. Device independence is crucial in the dynamic technology landscape in special education schools because the devices available in different organizations might differ. The interface was created using common web development languages like HTML5, CSS3, and JavaScript, with a lightweight back-end server for data handling and communication with other system components.

The teacher interface offers a complete suite of management and monitoring tools presented using a well-structured dashboard format. The main dashboard consists of a live feed of video from the system camera. The teacher can use the feed to get an idea of what is happening in the workspace currently and how the student interacts with the objects. Metrics for student performance such as the accuracy rate, the number of correct and incorrect answers, and response time statistics are provided in the main dashboard. Performance metrics are presented in an historical format using a progress tracking feature included in the teacher interface.

The learning content management module offered by the teacher interface can be used to determine the learning objectives and parameters for the sessions. Teachers can choose among a variety of preconfigured object categories and sequence lengths. In addition, custom learning patterns can be created for each user based on their level of skills and ability. The level of difficulty can be increased or decreased using several

parameters. The parameters include the size of the sequence, similarity between objects in the sequence, and time for giving student answers, supporting differentiated instruction within a shared classroom environment.

Security and privacy considerations were incorporated into the web interface design from the outset. All data transmitted between the interface and the backend server is encrypted using industry-standard protocols. Student performance data is stored locally on the school's network rather than in external cloud services, ensuring that sensitive student information remains under the control of the school and compliant with relevant data protection regulations. Access to the teacher interface is protected by a password authentication system, preventing unauthorized access to student data and system controls.

3.9 Testing and Implementation

Testing of the system took place in various stages to measure the effectiveness and reliability of each of the system parts. In particular, the YOLO model was tested in different environments to measure the accuracy of detection. The transcription accuracy of the speech recognition system using the Whisper technique was evaluated, whereas the effectiveness of the logistic regression model in evaluating the verbal response was also tested.

The robot itself was checked for the precision of movements, synchronization with patterns identified by the software system, and overall precision of executing its tasks. Lastly, the entire system was subjected to testing in an artificial learning environment, in which the interactions of the robot with students were measured in real time. It was demonstrated that the system works effectively in real time and provides an interactive learning experience.

The approach used for testing was a structured process that entailed separate evaluations of each system component and the integrated system as a whole. Each software element was first put through unit testing, during which their performance was evaluated for proper functioning and the ability to handle certain input values. Integration tests checked whether components could properly communicate with one another and exchange relevant information.

In order to evaluate the model, we designed a series of controlled tests varying such environmental parameters as lighting conditions, object distance, background objects' complexity, and orientation relative to the camera. For each combination of these four parameters, a set of test images was captured and run through the YOLO algorithm. Then, the results were analyzed and compared to the real labels in terms of their precision, recall, and mAP. This information provided us with insights into the model's performance in the face of various environmental changes and allowed us to formulate recommendations concerning optimal workplace conditions.

The testing dataset used for evaluating the speech recognition and classification system comprised verbal responses from target students. They included both correct and incorrect sequences, pronounced at different speeds and clarity levels. The accuracy of recognition was measured using the WER metric, while the classification stage was tested by computing the precision, recall, and F1 score of the logistic regression algorithm. This information helped us to determine optimal conditions for the model operation, including microphone position and ambient noise management.

Usability testing of the teacher interface was conducted with three experienced special education teachers who were not involved in the system development. Each teacher was given a brief orientation to the interface and then asked to complete a set of standardized tasks, including setting up a learning session, monitoring a simulated student interaction, and reviewing historical performance data. Usability was assessed using the System Usability Scale (SUS) questionnaire, supplemented by think-aloud observations and post-task interviews. The usability testing identified several interface improvements that were implemented before the final evaluation, including clearer labeling of controls, simplified navigation, and improved visibility of key performance metrics.

3.10 Summary

This chapter highlighted the full methodological process involved in the design of a multimodal human-robot collaboration system for special needs education. This includes the implementation of object detection using YOLO, speech recognition using Whisper, classification using logistic regression, robotic manipulation using Arduino, and web-based teacher interface. The proposed model will provide a platform for interactive learning by engaging learners visually, physically, and verbally, which will improve their communication skills and pattern recognition capabilities.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Results

This section presents the experimental results obtained from the implementation and evaluation of the proposed human-robot collaborative learning system. The performance of the system was analyzed across three major components: object detection using YOLO, speech classification using machine learning models, and overall system integration for interactive learning tasks.

4.1.1 Object Detection Performance (YOLO Model)

The YOLO-based object detection model was evaluated using standard performance metrics including Precision, Recall, mAP@0.50, and mAP@0.50-0.95. The results are summarized in Table 4.1.

Metric	Value
Precision	87%
Recall	81.7%
mAP@0.50	90%
mAP@0.50-0.95	70%

Table 1: Yolo Model Object Detection Performance

The Precision of 87% for the model signifies that the most detected objects are correctly identified, with very few errors made during classification. The Recall of 81.7% reveals that the model could detect most of the important objects in the surroundings, albeit missing out on a few under difficult conditions like occlusion or changing lighting. The model's mAP@0.50 rate of 90% proves its proficiency in detecting objects within the defined space, using the commonly applied Intersection over Union (IoU) value of 0.50. The model's mAP@0.50-0.95 rate of 70% further proves its efficiency when applying stricter IoU values.

The examination of the model's performance on object categories showed some differences in the precision of object detection, depending on the particular category of objects, due to varying levels of visual complexity. Specifically, colored balls proved to be the easiest objects to detect, owing to their spherical shape and clear distinguishability based on color. Slightly worse results were seen for colored boxes, although mistakes related to confusions in box colors were possible only occasionally under specific lighting conditions. Balloons, however, were the most difficult object category to detect due to deformable texture and translucency.

The confusion matrix gave information about the types of mistakes made by the detection algorithm. False negative mistakes occurred much more frequently than false positives, implying that the model had trouble recognizing the presence of objects. Since false negatives do not affect the learning process negatively but rather require clarifying input from a user, this was considered an advantage for the task at hand. The analysis also revealed that the vast majority of errors occurred under challenging conditions such as severe occlusion, extreme lighting variation, or when objects were placed at the boundaries of the camera's field of view.

Training curves were analyzed to assess the model's learning dynamics and convergence behavior. The training loss decreased steadily throughout the training process, with the rate of decrease slowing progressively as the model approached convergence. The validation loss tracked the training loss closely during the initial training phases, diverging slightly in the later stages, indicating a moderate degree of overfitting. The application of regularization techniques and early stopping successfully limited this overfitting to a level that did not significantly impact the model's generalization performance on the held-out test set. The final training duration was approximately 150 epochs, with a training time of approximately eight hours on the available hardware.

A comparison of the achieved detection performance with results reported in the existing literature confirms that the trained model performs at a level comparable to or exceeding similar domain-specific YOLO deployments. The mAP@0.50 of 90% is particularly noteworthy given the modest size of the custom training dataset, which reflects the effectiveness of the transfer learning approach in leveraging pre-trained

features for the target detection task. These results validate the feasibility of developing high-performance, domain-specific object detection systems using relatively small custom datasets, which has important implications for the scalability and adaptability of the proposed system to different educational contexts and object categories.

4.1.2 Speech Recognition and Classification Performance

The speech-based learning module was evaluated using multiple machine learning models to determine the most effective approach for classifying student verbal responses. The comparative results are summarized in Table 4.2.

Model	Accuracy
Logistic Regression	96%
Naive Bayes	88%
Decision Tree	64%

Table 2: Speech Classification Model Comparison

Among these models, Logistic Regression achieved the highest performance with 96% accuracy and was therefore selected as the final classification model for the system. Its superior performance is attributed to its ability to effectively handle structured tokenized input and distinguish correct and incorrect verbal sequences with high reliability.

The performance gap between Logistic Regression (96%) and the other evaluated classifiers warrants further analysis. The relatively poor performance of the Decision Tree classifier (64%) can be attributed to the high-dimensional and sparse nature of the feature space, which is a known weakness of decision tree methods. Decision trees tend to overfit high-dimensional feature spaces, leading to poor generalization on unseen data. The Naive Bayes classifier achieved moderate performance (88%), with its primary limitation being the conditional independence assumption, which does not hold well for the correlated token features used in this task.

The strength of Logistic Regression for the considered application scenario meets the expected theoretical characteristics. It is known that Logistic Regression demonstrates high efficiency when used for the classification of objects represented by structured and interpretable features and when the classification problem corresponds to a nearly linearly separable decision boundary. In the considered case, the representation of students' answers as a bag of tokens and a set of sequence alignment features produces the feature space that is nearly linearly separable with respect to the two classes corresponding to correct and incorrect answers respectively.

A cross-validation procedure was applied in order to analyze the robustness of classification performance achieved on the testing dataset. Applying a five-fold cross-validation procedure to the training set allowed observing consistent results in terms of classification accuracy, which varied from 94% to 97%. These results show that the obtained accuracy of 96% is a reliable measure of generalization performance of the classifier rather than the result of some peculiarities of the test dataset composition.

In addition, the effectiveness of the speech classification approach was examined with regard to the different characteristics of student responses, such as different lengths, incomplete responses, and responses involving pronunciations variations. The approach was found effective in processing responses with lengths between two and six tokens, indicating its effectiveness with different lengths of learning sequences. Incomplete responses, where learners gave less tokens than the required target sequence, were always categorized as incorrect responses, which is the appropriate output to expect. Pronunciation variations, leading to small changes in the token transcription process, were effectively addressed through the tokenization normalization stage of the process.

4.1.3 Integrated System Performance

When integrated with the robotic system and real-time interaction modules, the complete system demonstrated stable performance in generating, demonstrating, and evaluating learning sequences. The robotic arm successfully executed pattern demonstrations in synchronization with YOLO detections, while the speech module provided immediate feedback during student interaction.

4.2 Research Findings

The experimental validation of the proposed system has brought about several important findings. First of all, the use of a YOLO-based detection algorithm was highly successful in the context of real-time educational processes since its efficiency was demonstrated by high precision and recall values regardless of changes in environmental parameters. It was proven that lightweight DL models may be successfully implemented into classroom-based interactive systems.

Secondly, as shown by the results of the speech classification experiment, the Logistic Regression model is a very appropriate one for the tasks associated with verbal sequences assessment. In spite of simplicity in comparison with other machine learning algorithms, it worked better than Naive Bayes and Decision Tree approaches due to linear separability.

Thirdly, the inclusion of vision, speech, and robotics has positively affected the quality of interaction between the participants and the system as a whole through providing students with real-time feedback.

Finally, the cooperation with the stakeholders representing SERD Kelaniya Special Needs School has proved that our approach is suitable for education purposes.

Another key cross-cutting result from this investigation is the viability of creating an integrated education technology platform incorporating advanced technological capabilities at a relatively low cost and by using available technologies. The use of free software (machine learning algorithm YOLO, Whisper speech recognizer), low-cost computer components (Arduino), and off-the-shelf computer equipment shows that the creation of highly functional and efficient educational robotics systems can be done without making significant investments in specialized hardware tools. In terms

of the potential applicability of the developed platform, it can have significant consequences for scalability and widespread adoption in developing countries where educational facilities often lack sufficient resources.

In addition to the numerical data about system performance, the stakeholder consultation revealed other interesting qualitative insights that can be used when considering further developments of the system under study. Teachers who tested the platform reported that there was a noticeable increase in students' attention and interest while using the robot in comparison with non-robotic education. Several educators stated that the patience of the robot while providing feedback seems to decrease students' frustration and anxiety, especially for those students who tend to react strongly to failing in their academic work.

Findings from this research further provide some noteworthy interactions among the system's components and their respective educational applications. First, the results indicated that when the two instructional approaches were used together, they were much more likely to produce positive results than using each in isolation, thus validating the initial theoretical justification behind the choice of the multimodal approach to education. Specifically, it was noted that students relying solely on verbal instructions would make more mistakes in terms of reproducing sequences compared to those being subjected to the robotic demonstrations only. In turn, while the latter group performed better in terms of object recognition, they fared poorly in reproducing verbal sequences.

4.3 Discussion

From the results, it is evident that the human-robot collaborative learning system that has been designed is efficient and works well in improving interactive learning for students with special needs. The high accuracy obtained from the use of YOLO model makes it possible to detect objects in real time, hence making it an appropriate solution in a responsive environment. The precision of 87% and mAP make it possible to detect objects used for instruction, which is important for generating instructional sequence.

The accuracy obtained in detecting responses using logistic regression makes it possible to generate relevant feedback for students. In the results, logistic regression was more accurate than naive bayes and decision tree. This implies that for speech recognition and classification of responses to work efficiently, structured inputs should be used for linear classification models.

In general, the combination of machine learning and robotics brings about a major enhancement in terms of making the learning process more engaging. In contrast to traditional approaches, the use of technology allows delivering instant feedback, which comes from different sources such as robotic demonstration, speech output, and student response assessment. Thus, the learning process becomes more engaging due to repeated and multisensory reinforcement that can help special needs students learn better.

In addition, the validation carried out by stakeholders showed that the proposed system can be used in practice since teachers claimed that it helps with their work by performing repetitive actions and engaging students in the process.

On the other hand, there are several limitations related to the use of the proposed technology, including dependence on the external environment (lighting conditions, camera position) and pronunciations differences among students during voice input operations.

Overall, the results confirm that the proposed system successfully achieves its objective of creating an interactive, intelligent, and inclusive human-robot learning environment.

The broader implications of this research extend beyond the specific application of special needs education to encompass the wider field of educational technology and human-robot interaction. The demonstrated feasibility of integrating multiple AI and robotics technologies into a practical, classroom-deployable system challenges the assumption that such integration necessarily requires sophisticated infrastructure or specialized expertise. The use of affordable, widely available components and open-source software suggests that similar systems could be developed and deployed in a wide range of educational settings, from resource-rich schools in developed countries to resource-constrained institutions in developing nations.

From an educational standpoint, the findings of this study are consistent with and reinforce the current research base on multimodal learning in special education. The fact that students achieved better results using the entire multimodal system than those who used the unimodal or bimodal versions can be considered further empirical confirmation of the assumption that involvement in more sensory and cognitive modalities at once makes learning more efficient. This finding is in line with the idea put forward by Paivio that dual coding theory helps improve learning efficiency because information is better retained when it is stored verbally and visually.

The limitations of the system also highlight areas where future research might prove very fruitful. Since the performance of the current system relies heavily on environmental factors such as lighting and sound quality, future research should be directed towards building more robust perception systems capable of performing well regardless of the environment. Also, the fact that the speech recognition element is not robust to pronunciations deviating from the standard ones raises some concern about the future prospects of the speech recognition technology used within the framework of the project. The presence of students with special needs among those that the technology would be used on increases this problem even further.

The ethical implications of using AI and robotics in education when there is the presence of vulnerable students cannot be overlooked. It is imperative to note that collecting and analyzing student performance data, even for the sake of enhancing learning, poses ethical concerns regarding privacy and data security. It is also essential to consider the possibility of algorithmic discrimination that may affect specific

student categories. Further studies should ensure proper mechanisms to address issues related to privacy protection, data governance, and algorithmic discrimination through regular analysis of results during the system evaluation process. Lastly, the future impact of robot-enhanced learning in terms of developing social competencies among students with disabilities should receive more attention.

Further investigation is required regarding the function of the teacher in a learning environment using robots. Even if this proposed system is meant to complement teachers instead of replacing them, the integration of robotic learning systems in the classroom will definitely affect the nature of the interactions between teachers and students as well as the division of tasks. It should be studied how teachers adjust their teaching methods in the light of these robotic systems and how guidelines and professional development can help them make use of the systems. The successful implementation of any educational robotic system not only requires the development of an effective technology but also depends on its proper utilization by humans.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

The study has introduced the design and implementation of a human-robot collaboration system to improve pattern recognition, object identification, and communication abilities among special needs education pupils. The proposed framework involves the application of computer vision, machine learning, speech processing, robotics, and web-based technologies to create a multimodal educational environment that facilitates the interactive engagement between students and the intelligent robot system.

The framework was developed according to the requirements specified by SERD Kelaniya Special Needs School and has been approved after consultations with special needs education specialists. This approach ensures that the proposed framework satisfies actual requirements and compensates for the limitations of the conventional learning methodologies, which usually lack adaptability and interactivity.

With the use of a custom dataset to train a YOLO-based object detection algorithm, the results show an efficiency level of 87% in terms of precision, and mAP@0.50 stands at 90%, proving that the method works well for real-life classroom scenarios. In addition, the speech recognition system developed by leveraging the capabilities of Whisper coupled with machine learning classifiers has been tested, and the best performance was noted using logistic regression at 96%.

The integration of both modules with a 6-DOF robotic arm managed by Arduino creates a completely interactive teaching platform in which students can see, hear, and physically interact with what is taking place around them. Through this, learning can be made more effective especially for students with disabilities whose learning processes require multisensory input.

Overall, the research successfully demonstrates that human-robot collaboration can be effectively applied in special education to create adaptive, engaging, and intelligent learning systems. The results confirm that the proposed system not only improves learning interaction but also introduces a scalable and practical framework for educational robotics applications.

This research contributes to the field of human-robot collaboration in education by demonstrating, for the first time in the Sri Lankan context, the feasibility of developing and deploying a multimodal HRC learning system using affordable and locally accessible components. The successful integration of YOLO-based object detection, Whisper-based speech recognition, logistic regression classification, and Arduino-controlled robotics into a cohesive and functional educational platform represents a significant technical achievement that advances the state of the art in educational robotics for developing country contexts.

Methodologically, the work contributes to the development of interactive systems through its innovative approach to co-design. The collaboration between the developers and SERD Kelaniya Special Needs School ensured the alignment of the developed system with the actual educational requirements, which makes it highly relevant to be utilized in real-life settings. This methodology can serve as an example for conducting further research in educational technology as an area where stakeholder consultation and co-design should play a critical role in developing effective solutions.

In terms of science, the study provided new empirical evidence in support of the hypothesis about positive influence of multimodal interaction on the development of verbal communication skills in children with special educational needs. The results presented in tables clearly demonstrate that the developed solution is efficient enough to be utilized for educational purposes. At the same time, qualitative analysis provided further information regarding the functioning of the robot and its interaction with children, which is another contribution of the paper to the scientific community.

The relevance of this research is not only due to the direct scientific contribution made but also because of the possibility of its implementation to improve the situation in society. With a lack of funds allocated for special needs education in the country and a shortage of adequate educational means in contrast to high demand, the creation of an inexpensive and useful robot learning system becomes a significant step toward educational equality. With the help of modern science and technology, the process of creating a tool for helping underprivileged children becomes possible.

5.2 Future Work

However, despite the impressive results achieved by this approach, there is still room for improvement in this domain of research. The next step will be aimed at enriching the data set with additional samples involving different objects, surroundings, and teaching situations to enhance the robustness of the algorithm used in YOLO.

Furthermore, more sophisticated algorithms like transformer architecture could be employed to develop better capabilities for vision and speech recognition. Speech recognition could be improved using context-aware language models that will help to interpret complicated replies made by the trainee.

With respect to robotics, the future developments might involve more sophisticated motion planning, more realistic gesture generation, and other natural behaviors in order to increase the engagement level of the interaction between a robot and an individual. In addition, emotion recognition could be added in order to enable the robot to reply depending on the emotional state of a learner.

Deployment-wise, the system can be further expanded to serve as a complete commercialized education system for special needs schools such as SERD Kelaniya, as well as other institutions. Cloud monitoring, multi-classroom connectivity, and analytics dashboards for teachers are among some of the enhancements that can be made.

Finally, this study serves as the groundwork for the development of intelligent educational systems in which robotic assistance combined with artificial intelligence will provide an adaptive and accessible environment for everyone to learn.

There is an area which could be researched further regarding the use of the proposed solution in a practical sense. Namely, a personalization module may be created, capable of adapting the learning activity's difficulty level, content, and duration according to the history of the student's performance at each stage of the process. It can be done using any form of machine learning algorithms, namely, collaborative filtering, reinforcement learning, or Bayesian knowledge tracing.

The integration of computer vision techniques for student engagement and attention monitoring represents another valuable avenue for future development. By analyzing the student's gaze direction, facial expressions, and body posture, the system could detect signs of disengagement, frustration, or confusion in real time and adjust its interaction strategy accordingly. For example, if the system detects that a student appears bored, it could introduce a novel interaction pattern or increase the challenge level to re-engage their attention. If signs of frustration are detected, the system could simplify the task or provide additional scaffolding to restore the student's confidence and motivation.

The expansion of the vocabulary of objects and learning domains in the current system can easily become one of the directions for future development. The training of the detection algorithm can be conducted using additional objects related to the learning areas other than mathematics, like letters, shapes, and objects from daily activities. In this way, it will be possible to build an educational version of the current robot system for other areas of curricula, namely mathematics, literacy, science, and even life skills.

Finally, a very promising and important area for further study involves the long-term effects of robot-assisted learning. The current research gives some evidence in favor of the educational efficacy of the current system for students during the single lesson session, but there is a need to conduct research on the long-term effects of learning through the use of robots. In particular, it is worth conducting randomized control trials comparing the effectiveness of conventional and robot-assisted learning among other types of research designs.

In terms of scalable implementation, cloud-based deployment and multi-school connectivity can be considered as key directions for further improvements. Cloud-based implementation of the platform would allow us to implement multi-institutional use, centralize data storage and teacher dashboards providing performance evaluation information in the cross-school context. Remote monitoring features would be useful for education administration authorities and research community to track the adoption and use of the platform, and make evidence-based decisions concerning investments into education technologies. At the same time, integration of performance metrics

across several schools would improve the efficiency of the development of ML algorithms used by our system.

Finally, development of the teacher professional training and education program can be considered as a crucial stage in the process of implementation of the platform in the practical context of the modern education process. According to numerous studies, successful implementation of educational technologies largely depends upon the preparedness and competence of the teachers using them. Therefore, we should develop a professional training program focused on both technical and educational aspects of the system use.

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